Politecnico di Milano

School of Industrial and Information Engineering
M.Sc. in Aeronautical Engineering

ASSIGNMENT 2

Structural Dynamics and Aeroelasticity

Dynamic Aeroelastic Analysis of a Strut-Braced Wing with Tip Tank

Matteo Marwan Graziani – 307177

May 2026

Abstract

This report outlines the dynamic aeroelastic characterization of a cantilevered strut-braced wing carrying a tip fuel tank. The structural behavior is approximated through a rigorous Ritz–Galerkin spatial discretization. The study encompasses: the analytical derivation of the generalized mass, stiffness, and damping matrices; a modal convergence study to determine the optimal number of basis functions for the primary bending and torsional dynamics; the computation of the structural Frequency Response Function (FRF) to an aileron input; and a dynamic stability (flutter) analysis parameterized by the strut's chordwise attachment location. Finally, the system's aeroelastic response to both a deterministic step gust and a stochastic unit-intensity white-noise gust profile is evaluated. Based on the provided guidelines, the personal parameters adopted throughout are $L_1 = 18$ (letter R) and $L_2 = 1$ (letter A).

1 Problem Setup and Ritz–Galerkin Discretization

1.1 Kinematic Assumptions and Shape Functions

The analysis adopts the right-handed Cartesian coordinate system prescribed by the assignment guidelines. Its origin is situated on the elastic axis (EA) at the wing root. The positive axes are defined as follows: the spanwise axis (y) points towards the wingtip, the chordwise axis (x) points towards the trailing edge, and the vertical axis (z) points upwards. Consistent with this formulation, a positive twist angle (θ) and its associated torsional moment correspond to a pitch-up rotation.

The main wing structure is modeled as a uniform Euler–Bernoulli beam undergoing coupled flexural and torsional deformations. To convert the continuous partial differential equations into a discrete multi-degree-of-freedom system, the Ritz–Galerkin assumed-modes method is implemented. The transverse displacement of the elastic axis, $w(y,t)$, and the twist angle, $\theta(y,t)$, are expanded over sets of polynomial basis functions multiplied by their respective generalized coordinates:

$$w(y,t) = \underline{\mathbf{\Phi}}(y)\,\underline{\boldsymbol{q}}_w(t) = \sum_{i=1}^{N_w} \phi_i(y)\,q_{w,i}(t), \qquad \theta(y,t) = \underline{\mathbf{\Psi}}(y)\,\underline{\boldsymbol{q}}_\theta(t) = \sum_{j=1}^{N_\theta} \psi_j(y)\,q_{\theta,j}(t), \tag{1}$$

where N_w and N_θ denote the number of assumed modes for bending and torsion, respectively. To ensure rapid numerical convergence, the chosen shape functions must satisfy both the essential (kinematic) boundary conditions at the clamped root ($y = 0$) and the natural (dynamic) boundary conditions at the free tip ($y = b$). The essential boundary conditions require:

$$\phi_i(0) = 0, \qquad \phi_i'(0) = 0, \qquad \psi_j(0) = 0, \tag{2}$$

while the natural boundary conditions enforce the vanishing of the bending moment, shear force, and torsional moment at the free boundary:

$$\phi_i''(b) = 0, \qquad \phi_i'''(b) = 0, \qquad \psi_j'(b) = 0. \tag{3}$$

Imposing these six constraints yields the following polynomial basis functions:

$$\phi_i(y) = \left(\frac{y}{b}\right)^{i+1} - \frac{i(i+1)}{(i+2)(i+3)}\left(\frac{y}{b}\right)^{i+2} + \frac{i(i-1)}{(i+3)(i+4)}\left(\frac{y}{b}\right)^{i+3}, \tag{4}$$

$$\psi_j(y) = \left(\frac{y}{b}\right)^{j} - \frac{j}{j+1}\left(\frac{y}{b}\right)^{j+1}. \tag{5}$$

1.2 Derivatives of the Shape Functions

Evaluating the structural stiffness requires the second spatial derivative of the bending functions and the first spatial derivative of the torsional functions. Differentiating Eqs. (4) and (5) yields:

$$\phi_i''(y) = \frac{1}{b^2}\left[i(i+1)\left(\frac{y}{b}\right)^{i-1} - \frac{i(i+1)^2}{i+3}\left(\frac{y}{b}\right)^{i} + \frac{i(i-1)(i+2)}{i+4}\left(\frac{y}{b}\right)^{i+1}\right], \tag{6}$$

$$\psi_j'(y) = \frac{j}{b}\left[\left(\frac{y}{b}\right)^{j-1} - \left(\frac{y}{b}\right)^{j}\right]. \tag{7}$$

1.3 Strut and Tip Tank Parameters

The structural topology features an intermediate elastic constraint provided by a rigid strut connecting the wind-tunnel wall to the mid-span station $y_B = b/2$, positioned at an angle $\gamma = 15°$. The physical length of the strut, its equivalent transverse stiffness, and its default chordwise attachment coordinate are respectively given by:

$$L_{\text{strut}} = \frac{b/2}{\cos\gamma}, \qquad k_{\text{eq}} = \frac{EA}{L_{\text{strut}}}\sin^2\gamma, \qquad x_B = -0.5\,c. \tag{8}$$

The tip tank is idealized as a concentrated point mass. Its exact location is uniquely determined by the assigned parametric variables ($L_1 = 18$, $L_2 = 1$):

$$x_T = 0, \qquad y_T = \frac{L_1}{26}\,b, \qquad z_T = -1 - \frac{1}{2}\left(\frac{L_2}{26}\right). \tag{9}$$

The mass moment of inertia per unit span about the elastic axis is formulated via the parallel-axis theorem:

$$J_{EA} = J_G + m\,d^2, \tag{10}$$

where $d = 0.15\,c$ is the constant chordwise offset between the section's center of gravity (CG) and its elastic axis (EA).

2 Mass, Stiffness, and Damping Matrices

2.1 Mass Matrix

The generalized mass matrix is assembled using the virtual work of the D'Alembert inertial forces, accounting for both the distributed wing mass and the discrete tip tank. Assuming the tank is located at $x = 0$ and offset vertically by z_T, a twist angle θ induces a purely horizontal displacement $u_T = z_T\theta(y_T)$, while the vertical displacement remains $w_T = w(y_T)$. The virtual work of the inertial forces is thus expressed as:

$$\delta W_{\text{inertia}} = -\int_0^b \left[\delta w^T(m\ddot{w} - md\ddot{\theta}) + \delta\theta^T(J_{EA}\ddot{\theta} - md\ddot{w})\right] dy - M_T\delta w^T(y_T)\ddot{w}(y_T) - M_T z_T^2\delta\theta^T(y_T)\ddot{\theta}(y_T). \tag{11}$$

This formulation yields the symmetric generalized mass matrix:

$$\underline{\underline{\boldsymbol{M}}} = \begin{bmatrix} m\displaystyle\int_0^b \underline{\boldsymbol{\Phi}}^T\underline{\boldsymbol{\Phi}}\,dy + M_T\underline{\boldsymbol{\Phi}}(y_T)^T\underline{\boldsymbol{\Phi}}(y_T) & -md\displaystyle\int_0^b \underline{\boldsymbol{\Phi}}^T\underline{\boldsymbol{\Psi}}\,dy \\ -md\displaystyle\int_0^b \underline{\boldsymbol{\Psi}}^T\underline{\boldsymbol{\Phi}}\,dy & J_{EA}\displaystyle\int_0^b \underline{\boldsymbol{\Psi}}^T\underline{\boldsymbol{\Psi}}\,dy + M_T z_T^2\underline{\boldsymbol{\Psi}}(y_T)^T\underline{\boldsymbol{\Psi}}(y_T) \end{bmatrix}. \tag{12}$$

2.2 Stiffness Matrix

The internal virtual work comprises the flexural and torsional elastic strain energies, supplemented by the discrete potential energy stored in the strut. The total transverse displacement of the strut at the attachment point is linearly dependent on both bending and torsion:

$$w_B = w\left(\tfrac{b}{2}\right) - x_B\,\theta\left(\tfrac{b}{2}\right) = \underline{\boldsymbol{\Phi}}\left(\tfrac{b}{2}\right)\underline{\boldsymbol{q}}_w - x_B\,\underline{\boldsymbol{\Psi}}\left(\tfrac{b}{2}\right)\underline{\boldsymbol{q}}_\theta. \tag{13}$$

Consequently, the internal virtual work takes the form:

$$\delta W_{\text{int}} = \int_0^b \delta w''\,EJ\,w''\,dy + \int_0^b \delta\theta'\,GJ\,\theta'\,dy + \delta w_B\,k_{\text{eq}}\,w_B, \tag{14}$$

which defines the symmetric generalized structural stiffness matrix:

$$\underline{\underline{\boldsymbol{K}}} = \begin{bmatrix} EJ\displaystyle\int_0^b \underline{\boldsymbol{\Phi}}_{yy}^T\underline{\boldsymbol{\Phi}}_{yy}\,dy + k_{\text{eq}}\underline{\boldsymbol{\Phi}}\left(\tfrac{b}{2}\right)^T\underline{\boldsymbol{\Phi}}\left(\tfrac{b}{2}\right) & -x_B k_{\text{eq}}\underline{\boldsymbol{\Phi}}\left(\tfrac{b}{2}\right)^T\underline{\boldsymbol{\Psi}}\left(\tfrac{b}{2}\right) \\ -x_B k_{\text{eq}}\underline{\boldsymbol{\Psi}}\left(\tfrac{b}{2}\right)^T\underline{\boldsymbol{\Phi}}\left(\tfrac{b}{2}\right) & GJ\displaystyle\int_0^b \underline{\boldsymbol{\Psi}}_y^T\underline{\boldsymbol{\Psi}}_y\,dy + x_B^2 k_{\text{eq}}\underline{\boldsymbol{\Psi}}\left(\tfrac{b}{2}\right)^T\underline{\boldsymbol{\Psi}}\left(\tfrac{b}{2}\right) \end{bmatrix}. \tag{15}$$

Applying the Principle of Virtual Work ($\delta W_{\text{ext}} = \delta W_{\text{int}} - \delta W_{\text{inertia}}$) to the unloaded structure, the free-vibration equations of motion are recovered as $\underline{\underline{\boldsymbol{M}}}\ddot{\boldsymbol{q}} + \underline{\underline{\boldsymbol{K}}}\boldsymbol{q} = \underline{\boldsymbol{0}}$.

3 Free-Vibration Convergence Analysis (Point 1.a)

Before introducing the aerodynamic loads, the undamped free-vibration eigenvalue problem is solved to assess modal convergence:

$$\det\left(\underline{\underline{\boldsymbol{K}}} - \omega^2\underline{\underline{\boldsymbol{M}}}\right) = 0 \quad\Longrightarrow\quad \left[\underline{\boldsymbol{V}},\,\underline{\boldsymbol{\Lambda}}\right] = \text{eig}(\underline{\underline{\boldsymbol{K}}},\,\underline{\underline{\boldsymbol{M}}}), \qquad \omega_n = \sqrt{\Lambda_{nn}}. \tag{16}$$

The number of shape functions is iteratively increased until the relative variation in both the first bending and the first torsional natural frequencies drops below the prescribed tolerance:

$$\varepsilon = \left|\frac{\omega^{(N)} - \omega^{(N-1)}}{\omega^{(N)}}\right| < 10^{-3}. \tag{17}$$

The minimum threshold satisfying convergence simultaneously for both degrees of freedom dictates the truncation limits N_w and N_θ applied in all subsequent sections. Based on the numerical evaluation, convergence is achieved with $N_w = 4$ and $N_\theta = 8$.

4 Aerodynamic Matrices and Forcing Vector

4.1 Quasi-Steady Aerodynamic Model

The distributed aerodynamic loads are modeled adopting a strip-theory approach paired with a quasi-steady assumption. The effective angle of attack at the section's aerodynamic center is defined as:

$$\alpha(y,t) = \theta(y,t) - \frac{\dot{w}(y,t) + e\,\dot{\theta}(y,t)}{U_\infty}, \tag{18}$$

where $e = c/4$ is the constant chordwise offset between the aerodynamic center (AC) and the elastic axis (EA). The virtual work generated by the distributed aerodynamic lift and pitching moment translates to:

$$\delta W_{\text{aero}} = \int_0^b \delta w_{\text{AC}}\,q_\infty c\,C_{L/\alpha}\,\alpha(y,t)\,dy + \int_{b-b_a}^b \delta w_{\text{AC}}\,q_\infty c\,C_{L/\beta}\,\beta(t)\,dy + \int_{b-b_a}^b \delta\theta\,q_\infty c^2 C_{M_{\text{AC}}/\beta}\,\beta(t)\,dy, \tag{19}$$

where $w_{\mathrm{AC}} = w(y) + e\,\theta(y)$ and the dynamic pressure is $q_\infty = \frac{1}{2}\rho_\infty U_\infty^2$. Isolating the terms proportional to the states $\underline{\boldsymbol{q}}$ and their derivatives $\underline{\dot{\boldsymbol{q}}}$ yields the generalized aerodynamic stiffness and damping matrices:

$$\underline{\underline{\boldsymbol{K}}}_A = q_\infty c\, C_{L/\alpha} \begin{bmatrix} \underline{\underline{\mathbf{0}}} & \displaystyle\int_0^b \underline{\boldsymbol{\Phi}}^T \underline{\boldsymbol{\Psi}}\, dy \\ \underline{\underline{\mathbf{0}}} & e \displaystyle\int_0^b \underline{\boldsymbol{\Psi}}^T \underline{\boldsymbol{\Psi}}\, dy \end{bmatrix}, \tag{20}$$

$$\underline{\underline{\boldsymbol{C}}}_A = \frac{\rho_\infty U_\infty}{2}\, c\, C_{L/\alpha} \begin{bmatrix} \displaystyle\int_0^b \underline{\boldsymbol{\Phi}}^T \underline{\boldsymbol{\Phi}}\, dy & e \displaystyle\int_0^b \underline{\boldsymbol{\Phi}}^T \underline{\boldsymbol{\Psi}}\, dy \\ e \displaystyle\int_0^b \underline{\boldsymbol{\Psi}}^T \underline{\boldsymbol{\Phi}}\, dy & e^2 \displaystyle\int_0^b \underline{\boldsymbol{\Psi}}^T \underline{\boldsymbol{\Psi}}\, dy \end{bmatrix}. \tag{21}$$

4.2 Aileron Forcing Vector

An arbitrary aileron deflection $\beta(t)$ introduces an external generalized force vector, active exclusively over the spanwise extent b_a:

$$\underline{\boldsymbol{F}}_\beta = \begin{Bmatrix} q_\infty c\, C_{L/\beta} \displaystyle\int_{b-b_a}^b \underline{\boldsymbol{\Phi}}^T\, dy \\ q_\infty c\, C_{L/\beta}\, e \displaystyle\int_{b-b_a}^b \underline{\boldsymbol{\Psi}}^T\, dy + q_\infty c^2 C_{M_{\mathrm{AC}}/\beta} \displaystyle\int_{b-b_a}^b \underline{\boldsymbol{\Psi}}^T\, dy \end{Bmatrix}. \tag{22}$$

The complete, forced aeroelastic equations of motion are thereby governed by:

$$\underline{\underline{\boldsymbol{M}}}\, \underline{\ddot{\boldsymbol{q}}} + \underline{\underline{\boldsymbol{C}}}_A\, \underline{\dot{\boldsymbol{q}}} + \left(\underline{\underline{\boldsymbol{K}}} - \underline{\underline{\boldsymbol{K}}}_A\right) \underline{\boldsymbol{q}} = \underline{\boldsymbol{F}}_\beta\, \beta(t). \tag{23}$$

5 Frequency Response Function (Point 1.b)

To evaluate the system's dynamic susceptibility, a harmonic aileron input $\beta(t) = \beta_0\, e^{j\Omega t}$ is applied at a prescribed airspeed $U_\infty = 25\,\mathrm{m/s}$. Assuming a steady-state harmonic response $\underline{\boldsymbol{q}}(t) = \underline{\bar{\boldsymbol{q}}}\, e^{j\Omega t}$, Eq. (23) reduces to the frequency domain:

$$\left[-\Omega^2 \underline{\underline{\boldsymbol{M}}} + j\Omega \underline{\underline{\boldsymbol{C}}}_A + \left(\underline{\underline{\boldsymbol{K}}} - \underline{\underline{\boldsymbol{K}}}_A\right)\right] \underline{\bar{\boldsymbol{q}}} = \beta_0\, \underline{\boldsymbol{F}}_\beta. \tag{24}$$

The output of interest is the transverse displacement of point P, located at the leading edge of the wing tip ($y = b,\ x = -c/2$):

$$w_P = w(b) + \frac{c}{2}\,\theta(b) = \underline{\boldsymbol{\Phi}}(b)\, \underline{\boldsymbol{q}}_w + \frac{c}{2}\, \underline{\boldsymbol{\Psi}}(b)\, \underline{\boldsymbol{q}}_\theta =: \underline{\boldsymbol{H}}_P\, \underline{\bar{\boldsymbol{q}}}, \tag{25}$$

where $\underline{\boldsymbol{H}}_P = \left[\underline{\boldsymbol{\Phi}}(b),\ \frac{c}{2}\underline{\boldsymbol{\Psi}}(b)\right]$. The positive sign assigned to the torsional contribution term, $+\frac{c}{2}\theta(b)$, is strictly consistent with the adopted reference frame. Specifically, a positive pitch-up (nose-up) rotation is defined such that points shifting towards the leading edge undergo a positive vertical displacement. To maintain full kinematic alignment with the coordinate conventions prescribed in the problem statement, the cross-coupling term is uniquely defined with a positive sign, as identically mirrored within the MATLAB numerical implementation. The Frequency Response Function maps the aileron deflection to the tip displacement:

$$H_{w_P}(\Omega) = \frac{w_{P,0}}{\beta_0} = \underline{\boldsymbol{H}}_P \left[-\Omega^2 \underline{\underline{\boldsymbol{M}}} + j\Omega \underline{\underline{\boldsymbol{C}}}_A + \left(\underline{\underline{\boldsymbol{K}}} - \underline{\underline{\boldsymbol{K}}}_A\right)\right]^{-1} \underline{\boldsymbol{F}}_\beta. \tag{26}$$

The resulting amplitude and phase margins are subsequently analyzed via a Bode plot.

6 Flutter Analysis (Point 1.c)

6.1 State-Space Formulation

To rigorously assess aeroelastic stability limits, the homogeneous form of Eq. (23) ($\beta = 0$) is recast into a first-order state-space representation. Defining the state vector as $\underline{\boldsymbol{x}} = \{\underline{\boldsymbol{q}}^T,\ \dot{\underline{\boldsymbol{q}}}^T\}^T$, the system becomes:

$$\dot{\underline{\boldsymbol{x}}} = \underline{\underline{\boldsymbol{A}}}\,\underline{\boldsymbol{x}}, \qquad \underline{\underline{\boldsymbol{A}}} = \begin{bmatrix} \underline{\underline{\mathbf{0}}} & \underline{\underline{\boldsymbol{I}}} \\ -\underline{\underline{\boldsymbol{M}}}^{-1}\big(\underline{\underline{\boldsymbol{K}}} - \underline{\underline{\boldsymbol{K}}}_A\big) & -\underline{\underline{\boldsymbol{M}}}^{-1}\underline{\underline{\boldsymbol{C}}}_A \end{bmatrix}. \tag{27}$$

It is crucial to note that the structural stiffness $\underline{\underline{\boldsymbol{K}}}$ depends explicitly on the strut position x_B, whereas both $\underline{\underline{\boldsymbol{K}}}_A$ and $\underline{\underline{\boldsymbol{C}}}_A$ are functions of the freestream velocity U_∞.

6.2 Flutter Speed Determination

The complex eigenvalues $\lambda_k = \sigma_k \pm j\omega_k$ characterizing the dynamic matrix $\underline{\underline{\boldsymbol{A}}}$ are computed for progressively increasing values of U_∞. The critical flutter boundary V_F is identified when the real part of any conjugate pair crosses into the right-half plane (i.e., becomes positive), indicating dynamic instability:

$$V_F(x_B) = \min\big\{U_\infty \;\big|\; \exists\, i : \mathrm{Re}\big(\lambda_i(U_\infty, x_B)\big) > 0\big\}. \tag{28}$$

A parametric sweep over the applicable chordwise range accurately identifies the structural configuration that minimizes the onset flutter speed:

$$x_{B,\min} = \arg\min_{x_B}\ V_F(x_B). \tag{29}$$

7 Gust Response Analysis

7.1 Step Gust – Direct Recovery of Root Bending Moment (Point 1.d)

For transient analyses, the sub-critical cruise velocity is prescribed as $U_\infty = 0.75\, V_{F,\min}$. A uniform vertical gust velocity $v_G(t)$ perturbs the effective angle of attack according to the quasi-steady relation:

$$\alpha_G(t) = \frac{v_G(t)}{U_\infty}. \tag{30}$$

The aerodynamic load induced directly by this turbulence is distributed as $L_G(y,t) = q_\infty c\, C_{L/\alpha}\, \alpha_G(t)$.

$$\delta W_{aero} = \int_0^b \delta w_{AC} L_G(y,t)\, dy \tag{31}$$

Resolving this load yields the generalized gust force vector:

$$\underline{\boldsymbol{F}}_G = \frac{q_\infty c\, C_{L/\alpha}}{U_\infty} \begin{Bmatrix} \displaystyle\int_0^b \underline{\boldsymbol{\Phi}}^T\, dy \\ e\displaystyle\int_0^b \underline{\boldsymbol{\Psi}}^T\, dy \end{Bmatrix} = \frac{\rho_\infty U_\infty c\, C_{L/\alpha}}{2} \begin{Bmatrix} \displaystyle\int_0^b \underline{\boldsymbol{\Phi}}^T\, dy \\ e\displaystyle\int_0^b \underline{\boldsymbol{\Psi}}^T\, dy \end{Bmatrix}. \tag{32}$$

Assuming fixed control surfaces ($\beta = 0$), the operational state-space model is augmented with the gust disturbance:

$$\dot{\underline{\boldsymbol{x}}}(t) = \underline{\underline{\boldsymbol{A}}}\,\underline{\boldsymbol{x}}(t) + \underline{\underline{\boldsymbol{B}}}\, v_G(t), \qquad \underline{\underline{\boldsymbol{B}}} = \begin{Bmatrix} \underline{\mathbf{0}} \\ \underline{\underline{\boldsymbol{M}}}^{-1}\underline{\boldsymbol{F}}_G \end{Bmatrix}. \tag{33}$$

The primary output metric is the resultant bending moment at the cantilever root $M_x(0,t)$. Leveraging the Ritz expansion and the standard curvature–moment relation $M_x(0,t) = EJ\, w''(0,t)$:

$$M_x(0,t) = EJ\, w''(0,t) = EJ \sum_{i=1}^{N_w} \phi_i''(0)\, q_{w,i}(t). \tag{34}$$

Consequently, the output state matrix becomes:

$$\underline{\boldsymbol{C}}_{\text{state}} = \begin{bmatrix} EJ\, \underline{\boldsymbol{\phi}}_{yy}(0) & \underline{\mathbf{0}} \end{bmatrix}_{1\times 2(N_w+N_\theta)}, \tag{35}$$

with no direct feedthrough present ($\underline{\underline{\boldsymbol{D}}}_{\text{state}} = 0$). The transient history $M_x(0,t)$ to a strict step profile $v_G = \text{step}(t)$ is numerically integrated via a standard LTI solver.

7.2 Stochastic White-Noise Gust – Variance via Lyapunov (Point 1.e)

To evaluate fatigue and structural resilience, continuous atmospheric turbulence is modeled as a zero-mean, unit-intensity Gaussian white-noise process, $v_G(t) \sim \mathcal{WN}(0,1)$. The steady-state covariance matrix of the system state, defined as $\underline{\underline{\boldsymbol{P}}} = \mathbb{E}[\underline{\boldsymbol{x}}\underline{\boldsymbol{x}}^T]$, governs the algebraic Lyapunov equation:

$$\underline{\underline{\boldsymbol{A}}}\,\underline{\underline{\boldsymbol{P}}} + \underline{\underline{\boldsymbol{P}}}\,\underline{\underline{\boldsymbol{A}}}^T + \underline{\underline{\boldsymbol{B}}}\,\underline{\underline{\boldsymbol{B}}}^T = \underline{\underline{\mathbf{0}}}. \tag{36}$$

Upon solving Eq. (36) computationally using the MATLAB function $lyap(A, BB')$, the precise statistical variance of the root bending moment is directly extracted using the output matrix:

$$\sigma_{M_x}^2 = \underline{\boldsymbol{C}}_{\text{state}}\, \underline{\underline{\boldsymbol{P}}}\, \underline{\boldsymbol{C}}_{\text{state}}^T. \tag{37}$$

The feedthrough transmission matrix $\underline{\underline{\boldsymbol{D}}}$ is rigidly maintained at zero, as an ideal white-noise input possesses infinite energy; any direct non-zero input–output coupling would erroneously result in a mathematically unbounded variance for the structural response.